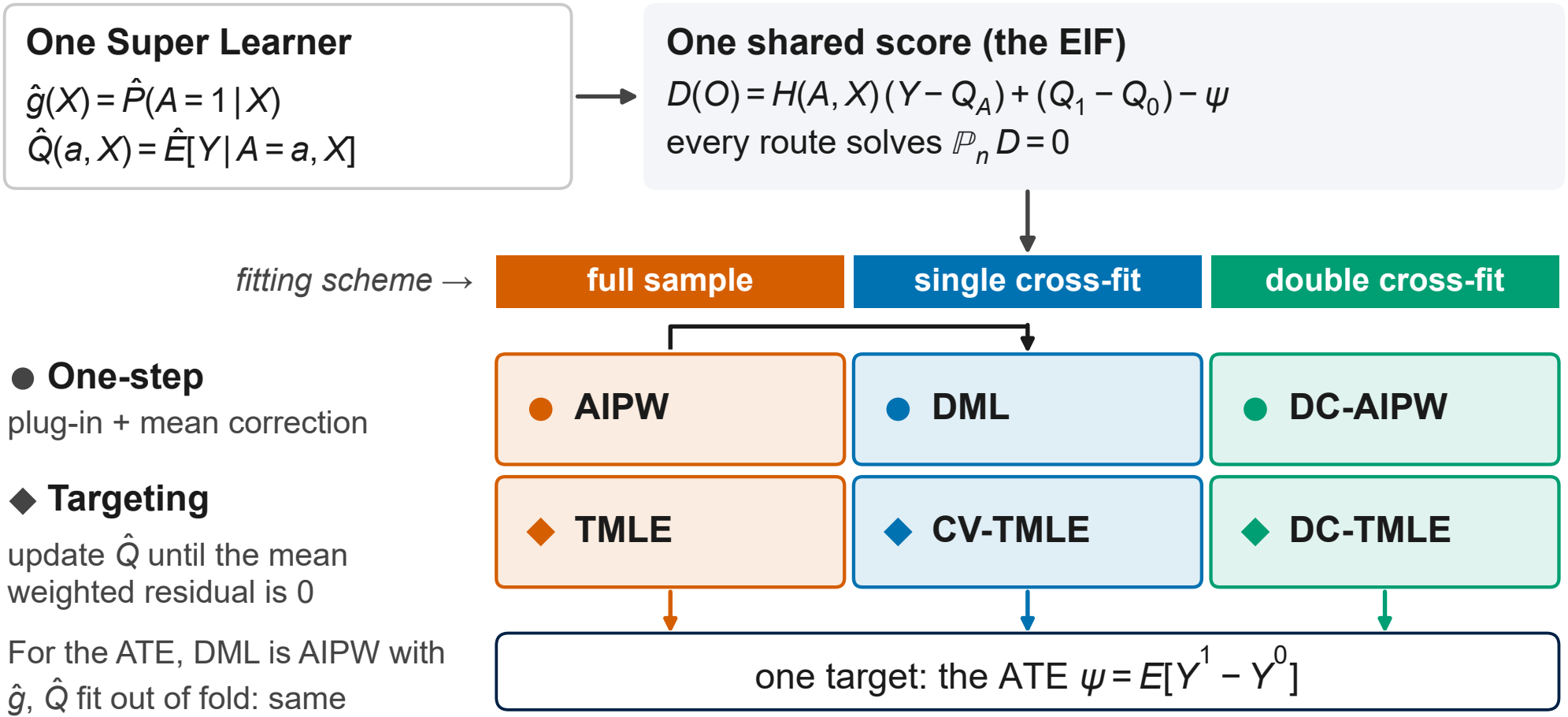

One Super Learner
$\hat{g}(X) = \hat{P}(A = 1 \mid X)$
$\hat{Q}(a, X) = \hat{E}[Y \mid A = a, X]$
One shared score (the EIF)
$D(O) = H(A, X)(Y - Q_A) + (Q_1 - Q_0) - \psi$
every route solves $\mathbb{P}_n D = 0$
fitting scheme →
full sample
single cross-fit
double cross-fit
● One-step
plug-in + mean correction
◆ Targeting
update $\hat{Q}$ until the mean weighted residual is 0
For the ATE, DML is AIPW with $\hat{g}$, $\hat{Q}$ fit out of fold: same row, next column. TMLE is the other row.
● AIPW
● DML
● DC-AIPW
◆ TMLE
◆ CV-TMLE
◆ DC-TMLE
one target: the ATE $\psi = E[Y^1 - Y^0]$